

Home Assignment 1

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A hidden Markov model for mobility tracking

Motion model

The target moves in $\mathbb{R}^2$ with state

$$\mathbf{X}_n = (X_n^1, \dot{X}_n^1, \ddot{X}_n^1, X_n^2, \dot{X}_n^2, \ddot{X}_n^2)^\top$$

The dynamics are

$$\mathbf{X}_{n+1} = \Phi \mathbf{X}_n + \Psi_z \mathbf{Z}_n + \Psi_w \mathbf{W}_{n+1} \quad (1)$$

where

- $\mathbf{Z}_n$ is a driving command (Markov chain with 5 states: stationary, east, north, west, south),
- $\mathbf{W}_{n+1} \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$ is Gaussian noise ($\sigma = 0.5$),
- Φ, Ψ_z, Ψ_w are built from $\Delta_t = 0.5$ and $\alpha = 0.6$.

Problem 1

1. Is $\{\mathbf{X}_n\}$ a Markov chain?

Answer: No.

Reasoning: A sequence is a Markov chain if it is "memoryless", meaning the future depends only on the current state, not the past.

Looking at equation (1), to calculate the next position $\mathbf{X}_{n+1}$, we need to know the current position $\mathbf{X}_n$ and the current driving command $\mathbf{Z}_n$. However, $\mathbf{Z}_n$ is not part of the vector $\mathbf{X}_n$. Because $\mathbf{Z}_n$ itself has "memory" (it follows its own Markov chain), knowing the past values of $\mathbf{X}$ would give us clues about $\mathbf{Z}_n$. Since the past provides extra information that the current $\mathbf{X}_n$ alone does not have, $\mathbf{X}_n$ is not a Markov chain.

2. Is $\tilde{\mathbf{X}}_n = (\mathbf{X}_n^\top, \mathbf{Z}_n^\top)^\top$ a Markov chain?

Answer: Yes.

Reasoning: When we combine $\mathbf{X}_n$ and $\mathbf{Z}_n$ into one "augmented" state vector $\tilde{\mathbf{X}}_n$, we now have all the information needed:

- For $\mathbf{X}$: The next state $\mathbf{X}_{n+1}$ depends only on the current $\mathbf{X}_n$ and $\mathbf{Z}_n$, and the independent noise $\mathbf{W}_{n+1}$.

- For $\mathbf{Z}$: The next command $\mathbf{Z}_{n+1}$ depends only on the current $\mathbf{Z}_n$ (as it is a Markov chain). Since the future state $\tilde{\mathbf{X}}_{n+1}$ depends only on the current state $\tilde{\mathbf{X}}_n$, it satisfies the Markov property.

3. Does the trajectory look like a reasonable trajectory of a moving target?

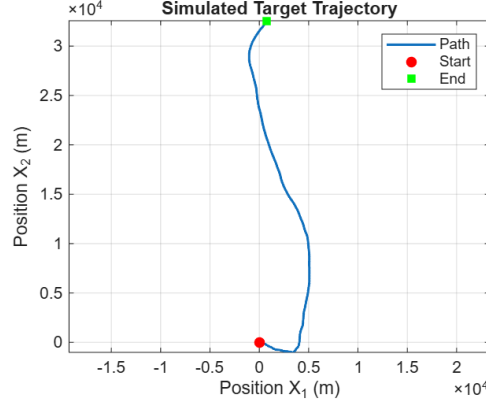


Figure 1: Simulated Target Trajectory

Answer: Yes.

Reasoning:

1. Smoothness: Because the model includes velocity and acceleration, the target doesn't teleport or jump erratically. It has "inertia," meaning it moves smoothly from one point to the next.

2. Purposeful Motion: The transition matrix P has a very high diagonal value ($16/20 = 0.8$). This means that if the driver starts moving "North," they will likely keep moving North for a while before changing direction. This creates long, clear paths rather than messy, random zig-zags.

3. Realism: The combination of steady driving commands (Z_n) and small random vibrations (W_n) mimics how a real vehicle or person moves—mostly following a path but with slight natural deviations.

Observation model

Problem 2

1. Why $\{(\tilde{\mathbf{X}}_n, \mathbf{Y}_n)\}_{n \in \mathbb{N}}$ form a Hidden Markov Model (HMM)

A Hidden Markov Model consists of two components:

(1) The hidden state must be a Markov Chain. In Problem 1, we already established that the augmented state vector $\tilde{\mathbf{X}}_n = (\mathbf{X}_n^\top, \mathbf{Z}_n^\top)^\top$ is a Markov chain:

$$p(\tilde{\mathbf{X}}_{n+1} \mid \tilde{\mathbf{X}}_0, \dots, \tilde{\mathbf{X}}_n) = p(\tilde{\mathbf{X}}_{n+1} \mid \tilde{\mathbf{X}}_n).$$

(2) The observation must depend only on the current hidden state. Looking at the observation model:

$$Y_n^\ell = v - 10\eta \log_{10} \|(X_n^1, X_n^2)^\top - \boldsymbol{\pi}_\ell\| + V_n^\ell \quad (2)$$

In this equation, to calculate the received signal strength $\mathbf{Y}_n$ at time n , we only need the target's current position (X_n^1, X_n^2) , which is entirely contained within the current state $\tilde{\mathbf{X}}_n$. The noise V_n^ℓ , which is completely independent of all past states and past observations. Therefore:

$$p(\mathbf{Y}_n \mid \tilde{\mathbf{X}}_0, \dots, \tilde{\mathbf{X}}_n, \mathbf{Y}_0, \dots, \mathbf{Y}_{n-1}) = p(\mathbf{Y}_n \mid \tilde{\mathbf{X}}_n).$$

This satisfies the definition of an HMM.

2. Transition density $p(\mathbf{y}_n \mid \tilde{\mathbf{x}}_n)$

Given a known state vector $\tilde{\mathbf{x}}_n$, the target's position (x_n^1, x_n^2) is deterministic. Let $\mu_{n,\ell}$ denote the expected, noise-free signal strength from the ℓ -th base station:

$$\mu_{n,\ell} = v - 10\eta \log_{10} \|(x_n^1, x_n^2)^\top - \boldsymbol{\pi}_\ell\|$$

Based on equation (2), the observed signal Y_n^ℓ is the sum of this deterministic value $\mu_{n,\ell}$ and the noise V_n^ℓ . Since $V_n^\ell \sim N(0, \varsigma^2)$, the conditional distribution of a single observation Y_n^ℓ is Gaussian:

$$Y_n^\ell \mid \tilde{\mathbf{x}}_n \sim N(\mu_{n,\ell}, \varsigma^2)$$

The corresponding probability density function (PDF) is:

$$p(y_n^\ell \mid \tilde{\mathbf{x}}_n) = \frac{1}{\sqrt{2\pi\varsigma^2}} \exp\left(-\frac{(y_n^\ell - \mu_{n,\ell})^2}{2\varsigma^2}\right)$$

The noise variables $\{V_n^\ell\}_{\ell=1}^s$ are independent. Consequently, given the state $\tilde{\mathbf{x}}_n$, the observations from the $s = 6$ base stations are conditionally independent. The joint transition density is the product of the marginal densities:

$$p(\mathbf{y}_n \mid \tilde{\mathbf{x}}_n) = \prod_{\ell=1}^6 p(y_n^\ell \mid \tilde{\mathbf{x}}_n)$$

Substituting the individual PDFs into the product, we obtain the final transition density function:

$$p(\mathbf{y}_n \mid \tilde{\mathbf{x}}_n) = \frac{1}{(2\pi\varsigma^2)^3} \exp\left(-\frac{1}{2\varsigma^2} \sum_{\ell=1}^6 (y_n^\ell - \mu_{n,\ell})^2\right)$$

Expanding $\mu_{n,\ell}$ with its full expression yields:

$$p(\mathbf{y}_n \mid \tilde{\mathbf{x}}_n) = \frac{1}{(2\pi\varsigma^2)^3} \exp\left(-\frac{1}{2\varsigma^2} \sum_{\ell=1}^6 (y_n^\ell - v + 10\eta \log_{10} \|(x_n^1, x_n^2)^\top - \boldsymbol{\pi}_\ell\|)^2\right)$$

Substituting the given parameters ($v = 90$, $\eta = 3$, and $\varsigma = 1.5$), we obtain:

$$p(\mathbf{y}_n \mid \tilde{\mathbf{x}}_n) = \frac{1}{(4.5\pi)^3} \exp\left(-\frac{2}{9} \sum_{\ell=1}^6 (y_n^\ell - 90 + 30 \log_{10} \|(x_n^1, x_n^2)^\top - \boldsymbol{\pi}_\ell\|)^2\right)$$

Mobility tracking using SMC methods

Problem 3

The general sequential weight update equation for a particle filter is given by:

$$w_n^i = w_{n-1}^i \frac{p(\mathbf{y}_n \mid \tilde{\mathbf{x}}_n^i) p(\tilde{\mathbf{x}}_n^i \mid \tilde{\mathbf{x}}_{n-1}^i)}{q(\tilde{\mathbf{x}}_n^i \mid \tilde{\mathbf{x}}_{n-1}^i, \mathbf{y}_n)}$$

The specific proposal density:

$$q(\tilde{\mathbf{x}}_n^i \mid \tilde{\mathbf{x}}_{n-1}^i, \mathbf{y}_n) = p(\tilde{\mathbf{x}}_n^i \mid \tilde{\mathbf{x}}_{n-1}^i)$$

By substituting this proposal density into the denominator of the weight update equation, the transition density term $p(\tilde{\mathbf{x}}_n^i | \tilde{\mathbf{x}}_{n-1}^i)$ elegantly cancels out from both the numerator and the denominator:

$$w_n^i = w_{n-1}^i \frac{p(\mathbf{y}_n | \tilde{\mathbf{x}}_n^i) p(\tilde{\mathbf{x}}_n^i | \tilde{\mathbf{x}}_{n-1}^i)}{p(\tilde{\mathbf{x}}_n^i | \tilde{\mathbf{x}}_{n-1}^i)}$$

This yields the simplified weight update formula:

$$w_n^i = w_{n-1}^i p(\mathbf{y}_n | \tilde{\mathbf{x}}_n^i)$$

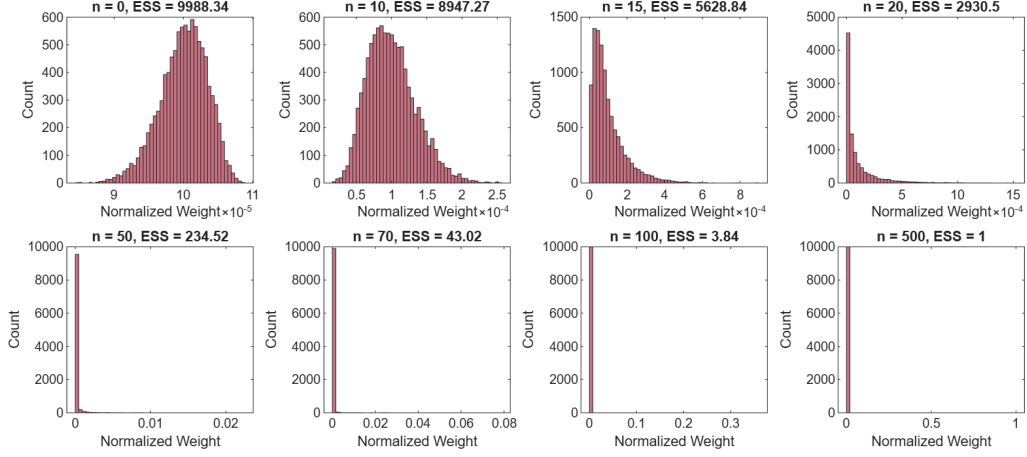


Figure 2: Weight Distribution and ESS Statistics

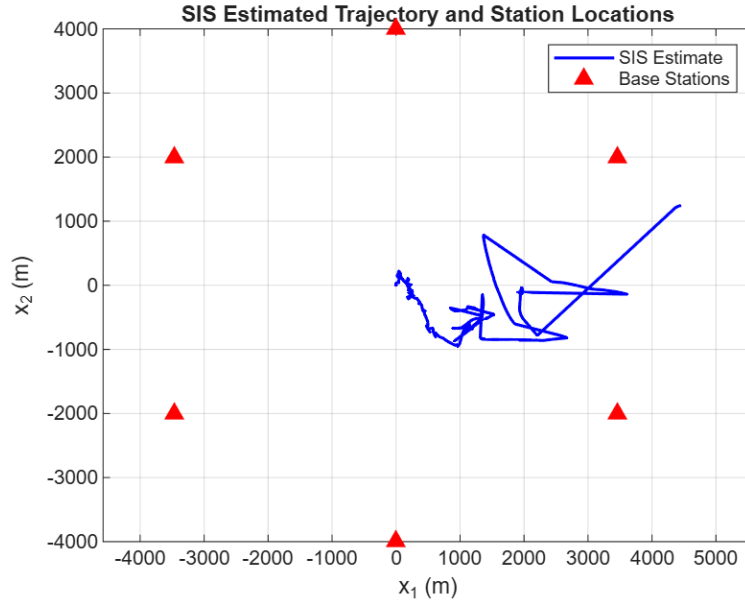


Figure 3: SIS Filter Trajectory Estimate

Conclusions:

1. Weight degeneracy:

- Initial Stability ($n = 0$ to $n = 10$): At the start, the ESS remains high (9988.34 at $n = 0$ and 8947.27 at $n = 10$). This suggests that the initial particles are well-distributed and the early observations are highly consistent with the state transitions.

- Accelerated Decay ($n = 15$ to $n = 70$): The filter begins to struggle mid-way. The *ESS* drops by nearly half between $n = 10$ and $n = 15$ (5628.84), and collapses further to 43.02 by $n = 70$. This indicates that as the trajectory grows longer, fewer particles can "keep up" with the cumulative likelihood of the RSSI measurements.
- Total Degeneracy ($n \geq 100$): By $n = 100$, the *ESS* hits a critical low of 3.84, and by $n = 500$, it reaches the theoretical minimum of 1.0. The histograms confirm this: by the final step, a single particle dominates the entire distribution with a weight of 1, while the other 9,999 particles are effectively ignored.

2. Impact on trajectory estimation:

The trajectory in Fig.3 reflects this loss of diversity. While the estimate might remain somewhat stable during the first 50 steps (where *ESS* is still relatively high), it becomes increasingly erratic as n approaches 500. Since the estimate $\hat{X}_n$ is a weighted average, once the *ESS* reaches 1, the "estimated" path is simply the random walk of one surviving particle, which may have drifted significantly from the true target location.

3. Necessity of resampling:

The experiment demonstrates that the pure SIS algorithm is impractical for processing long sequences of observations. To resolve the degeneracy issue, a Resampling step (SISR) is essential. Resampling restores sample diversity and filter effectiveness by duplicating high-weight particles and discarding those with negligible weights.

Problem 4

The Sequential Importance Sampling with Resampling (SISR) algorithm is an improvement of the SIS method. As discussed in the previous section, a major drawback of SIS is the weight degeneracy. SISR addresses this by introducing a resampling step. It eliminates particles with small weights and replicate those with relatively large weights, thereby maintaining a more balanced and representative particle set.

After implementation of SISR, the estimated trajectory of the object is shown as Fig.4. The most probable driving command is inferred by performing a weighted vote across all particle samples for each step, and the detailed result is demonstrated clearly in Fig. 5.

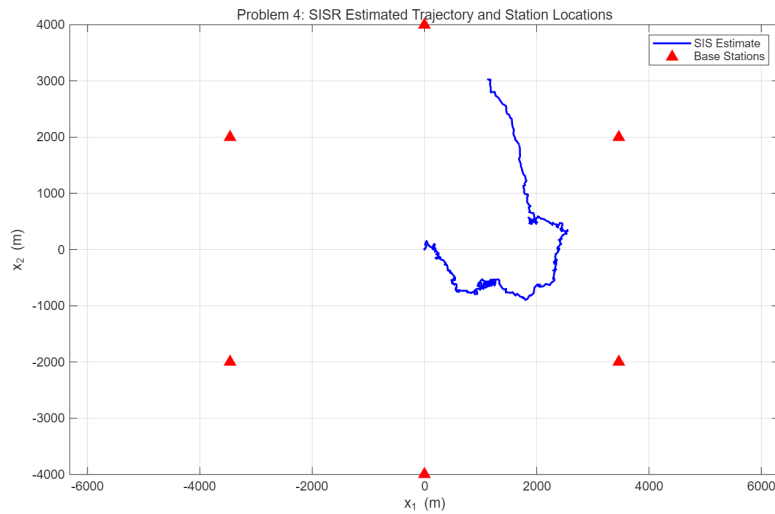


Figure 4: Position Estimation with SISR

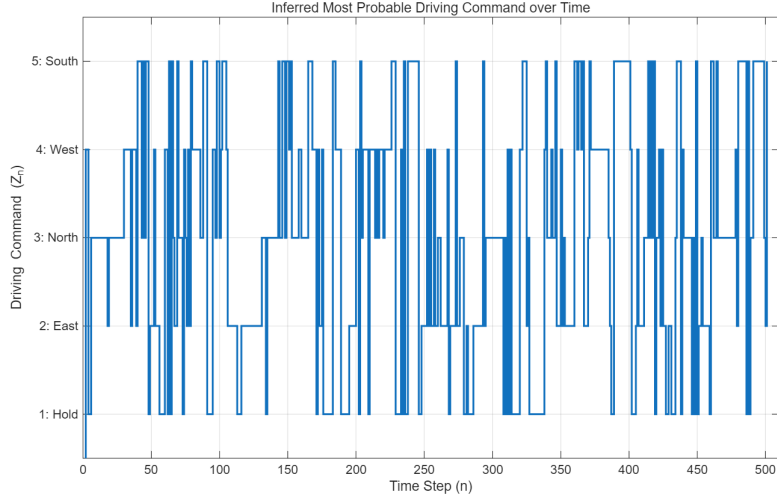


Figure 5: Inferred Driving Command

Conclusions:

1. **Coherence of the Estimated Trajectory:** The SISR algorithm yields a highly stable and coherent trajectory estimation. While the estimated path exhibits minor local stochastic fluctuations due to process noise and sudden shifts, the overall trajectory remains distinct and plausible.
2. **Effectiveness of the Resampling Mechanism:** The better trajectory reflects that resampling strategy ensures a healthy and informative particle pool and guarantees the robustness of long-term tracking.
3. **Alignment of Inferred Driving Command:** The inferred most probable driving commands aligns well with the underlying prior Markov model. Fig.5 indicates that the line segments vary in both length and width, with a high proportion of horizontal segments. This aligns with the transition matrix design, where maintaining the current direction has an 80% probability, while transitions to other four status each occur with a 5% probability.

SMC-based model calibration

Problem 5

In this section, the standard deviation ς of noise(Y_n^l) is unknown but be bounded within the interval $(0,3)$. To calibrate this parameter, we perform a grid search over ς to maximize the log-likelihood of the observation sequence. However, the exact marginal likelihood is analytically intractable because there's no closed-form solution to it. But with SISR, the likelihood at each time step can be approximated as,

$$p(y_n|y_{0:n-1}, \varsigma) \approx \sum_{i=1}^N \tilde{w}_n^{(i)}$$

By aggregating the logarithmic sum of these $\tilde{w}_n^{(i)}$ across all m time steps, we effectively construct a pointwise approximation of the total log-likelihood, which is $\ell_m^N(\varsigma_j)$. For every possible value of ς_j , we calculate this approximation and select the parameter that yields the maximum value as our optimal $\hat{\varsigma}_m$.

Changes of approximate log-likelihood as ς increases is shown in Fig.6. And the $\hat{\varsigma}_m$ is selected to be 2.2.

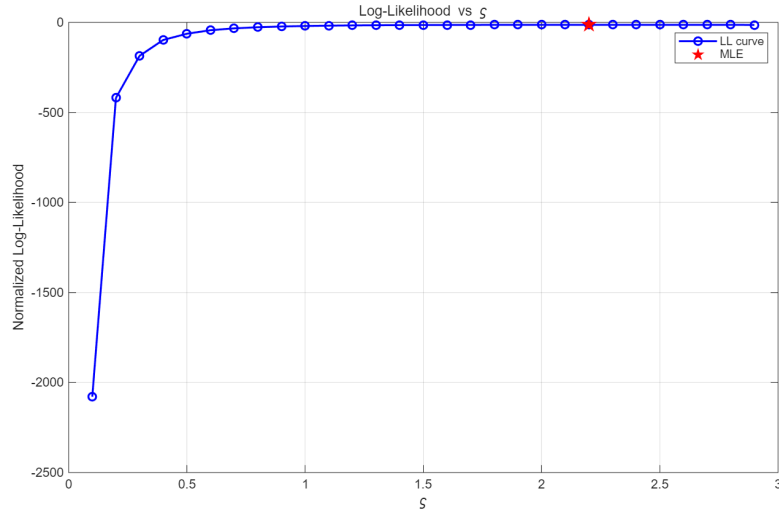


Figure 6: Grid Search for best ς

Substituting $\hat{\varsigma}_m = 2.2$ into algorithm SISR yields an estimated trajectory as in Fig.7.

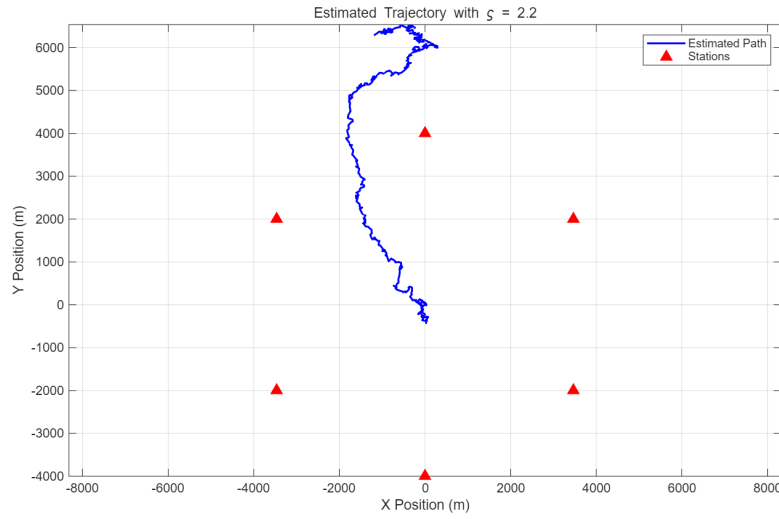


Figure 7: Trajectory under best ς